

Nikhil Gangaramani

AI Engineer — RAG Systems & High-Throughput Data Pipelines

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SUMMARY

AI engineer focused on production-grade retrieval systems and the data infrastructure underneath them: RAG pipelines over SEC filings, ingestion/routing layers moving 40–100M data points per day, and query layers rebuilt for speed (+75% on Simfund/MarketPulse). Comfortable owning systems end-to-end — from chunking strategy and vector-index hygiene to Kafka partitioning and PostgreSQL query plans.

EXPERIENCE

Institutional Shareholder Services (ISS)

2023 — Present

Software Engineer — Data & AI · Simfund (MarketPulse)

- Rebuilt the hot query path on the Simfund (MarketPulse) fund-analytics platform — covering indexes, denormalized read models, and aggregate pushdown — for a **+75% query performance increase**.
- Built Golang ingestion services handling tens of millions of market data points daily with strict ordering and idempotency guarantees.
- Shipped RAG tooling over SEC filings so analysts query institutional holdings in natural language instead of SQL.

Backend Platform Work

2022 — 2023

Software Engineer · JVM Services

- Owned latency-sensitive Java services; profiled allocation pressure and tuned generational GC to cut p99 pauses.
- Introduced structured concurrency and load-shedding so services degrade predictably under burst load.

PROJECTS

MI Buddy — RAG over SEC 13F institutional filings

- Parse → chunk → embed → retrieve/rerank pipeline; streamed answers with citations back into the source filing. (Python, pgvector, FastAPI, LLM APIs)

High-Throughput Routing — 40–100M data points/day

- Golang worker pools fanning Kafka partitions through a rules-based router; backpressure + dead-letter handling; at-least-once delivery into idempotent sinks.

SKILLS

Languages: Golang, Python, Java, TypeScript, SQL

AI/Retrieval: RAG architecture, embeddings, pgvector/FAISS, LLM orchestration & evals

Data/Infra: Kafka, PostgreSQL, Redis, Docker/Kubernetes, AWS